

## TARGETING PROGRAMMED CELL DEATH IN NEURODEGENERATIVE DISEASES

Miquel Vila\* and Serge Przedborski\*<sup>‡§</sup>

Molecular pathways of programmed cell death (PCD) are activated in various neurodegenerative disorders including Parkinson's disease, amyotrophic lateral sclerosis and Huntington's disease. In these diseases, PCD might be pathogenic, and targeting it might mitigate neurodegeneration. To identify potential neuroprotective targets within the PCD machinery, the expression and activity of some of its components have been altered by genetic or pharmacological means in experimental models of neurodegenerative diseases. The results of these studies have provided leads for the development of neuroprotective strategies for these progressive, disabling and often fatal disorders.

### NEUROLOGICAL DISEASES

**TUNEL TECHNIQUE**  
This technique enables the visualization of cells undergoing apoptosis by labelling the broken ends of the double-stranded DNA with biotin-conjugated dUTP, using the enzyme terminal deoxynucleotidyl transferase.

Programmed cell death (PCD) is a physiological process in which molecular programs that are intrinsic to the cell are activated to cause its destruction. This process is a fundamental property of all multicellular organisms and is crucial for their development, for organ morphogenesis, for tissue homeostasis and for defense against infected or damaged cells. The importance of PCD is emphasized by its remarkable degree of evolutionary conservation. However, excessive PCD can cause unwarranted cell death, which might lead to diseases such as immunodeficiency and neurodegeneration.

The term PCD is often used interchangeably with 'apoptosis' — a morphological form of cell death that is characterized by membrane blebbing, shrinkage of the cell body, nuclear condensation and DNA fragmentation. However, apoptosis is only one morphological form of PCD<sup>1</sup>; the molecular pathways linked to PCD are implicated in cell-death processes, the morphological diversity of which extends beyond apoptosis<sup>2,3</sup>. At times, even necrosis, which is traditionally considered to be a 'passive' death process (that is, death that does not rely on intracellular signalling pathways), has been prevented by anti-PCD compounds<sup>4</sup>. In this article, we group under the term PCD (also sometimes referred to as 'active cell death'<sup>5</sup>) all

cell death forms that involve active intracellular processes, and use 'apoptosis' only in reference to the morphology of dying cells.

Over the past ten years, three experimental waves have characterized the study of PCD in neurodegeneration. Initially, the focus was on the search for apoptotic cells in *post mortem* tissues. This effort was undertaken in relation to several neurodegenerative disorders and gave rise to conflicting results. One lesson learned from these investigations was that looking for apoptosis in *post mortem* human tissue is complicated by many conceptual and technical factors. First, it is difficult to detect apoptosis owing to the presumed low daily rate of neuron loss in neurodegenerative disorders and the presumed rapid disappearance of apoptotic cells. Second, *post mortem* specimens typically derive from advanced stages of the disease, when most of the neurons that are affected by the pathological process are already lost. Third, most morphological *post mortem* studies rely on the so-called terminal deoxynucleotidyl transferase-mediated dUTP nick-end labelling (TUNEL) technique to document the presence of apoptotic cells. However, we now know that TUNEL is not specific for apoptosis, especially in human *post mortem* tissue, in which factors such as hypoxia can produce TUNEL-positive, non-apoptotic DNA damage<sup>6</sup>.

\*Department of Neurology,  
‡Department of Pathology,  
and §Center for Neurobiology  
and Behavior, Columbia  
University, 650 West 168th  
Street, BB-307, New York,  
New York 10032, USA.  
Correspondence to M.V.  
e-mail:  
MV199@columbia.edu.  
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## Box 1 | Programmed cell death and Alzheimer's disease

Amyloid- $\beta$  (A $\beta$ ) — a peptide that accumulates in the brains of people with Alzheimer's disease (AD) and forms amyloid plaques — directly induces apoptosis of cultured neurons<sup>143</sup>. Therefore, many researchers have looked for signs of programmed cell death (PCD) in Alzheimer patients. Along this line, fragmentation of nuclear DNA, detected by terminal deoxynucleotidyl transferase-mediated dUTP nick-end labelling (TUNEL)<sup>144</sup>, has been detected in brains of people with AD, although this technique lacks specificity. More specific biochemical evidence that PCD might occur in AD is provided by the detection of activated caspases 3, 8 and 9 in hippocampal neurons of brains affected by AD<sup>145–148</sup>. Moreover, pharmacological or molecular inhibition of particular members of the caspase family, such as caspases 2, 3, 8 and 12, has been reported to offer partial or complete protection against A $\beta$ -induced apoptotic cell death *in vitro*<sup>25,149–151</sup>. A $\beta$  is derived from  $\gamma$ -secretase-mediated processing of the amyloid precursor protein (APP), but it has been shown, in both cultured cells and brains affected by AD, that APP can also be cleaved by caspases, such as caspase 3, at sites distinct from the classic secretase-processing sites<sup>152</sup>. Caspase-mediated cleavage of APP not only releases A $\beta$ , but can also release a carboxy-terminal peptide that is a potent inducer of apoptosis<sup>153</sup>. Similarly, caspase 3-cleaved fragments of tau, a microtubule-associated protein that is the primary protein component of the filaments found in the brains of people with AD, have also been detected in *post mortem* samples<sup>148</sup>. Despite all of these data, there is no evidence of caspase activation or apoptotic cell death in animal models of AD and, therefore, there is no evidence *in vivo* for a potential beneficial effect of blocking PCD pathways in AD.

Given these problems, many investigators have stopped using an exclusively morphological approach, and now include techniques that assess molecular components of the PCD machinery. Although this combined approach has often shed light on the state of PCD in neurodegenerative diseases, none of these *post mortem* findings have established a role for PCD in the pathogenic process.

More recently, investigators have lost interest in the demonstration of PCD-associated cellular and molecular changes in human tissues, and they now focus on showing the actual role of PCD in the neurodegenerative process. To achieve this goal, key PCD molecules have been manipulated (either in transgenic or knockout studies), or inhibited by pharmacological agents or viral vectors in experimental models of neurodegenerative disorders. These studies have not only identified PCD components that either promote or prevent neuronal death, but they have also disclosed molecular targets for the development of drugs for preventing and treating neurodegenerative disorders.

In this article, after discussing the molecular composition of the PCD machinery, we review the results of manipulating such molecular pathways on the fate of neurons in experimental models of three prominent neurodegenerative disorders: **Parkinson's disease** (PD), **amyotrophic lateral sclerosis** (ALS) and **Huntington's disease** (HD). Although some work has been done on PCD in Alzheimer's disease<sup>7</sup> (BOX 1), the benefit of targeting PCD in experimental models of Alzheimer's disease remains to be determined; so, we will not discuss Alzheimer's disease here. Similarly, we will not discuss PCD in stroke<sup>8</sup>, as this disorder falls outside the scope of an article on neurodegenerative diseases<sup>9</sup>.

## ZymoGENS

The inactive precursors of enzymes — often transformed into the active enzyme by partial proteolysis.

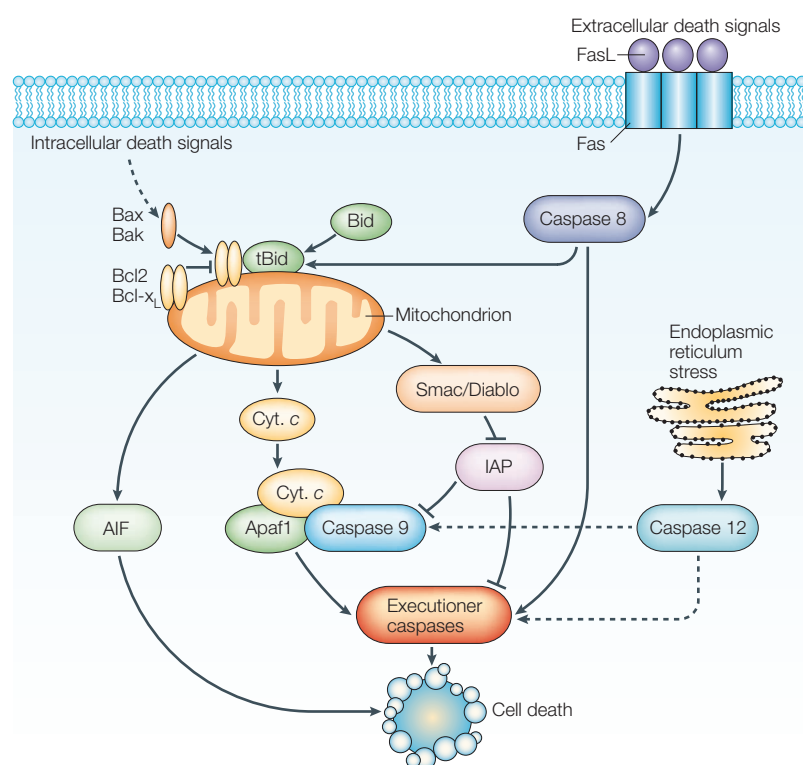
## Molecular pathways of PCD

The key mediators of PCD are the proteolytic enzymes called 'caspases', which cleave their substrates after specific aspartic acid residues. Caspases exist as ZymoGENS (procaspases) in almost every animal cell, and they become activated in response to intracellular signalling pathways that are triggered by various cellular perturbations, such as DNA damage and withdrawal of trophic support. The family of mammalian caspases comprises 15 members, which can be divided into initiators (procaspases 2, 8, 9 and 10) and effectors (procaspases 3, 6 and 7) of PCD.

The initiators are the first caspases to become activated in the PCD cascade. They have long amino-terminal prodomains that contain specific protein–protein interaction motifs. Through these domains, initiator **caspases 8** and **9** are activated after being aggregated by the adaptor molecules **FADD** (Fas-associating protein with death domain)<sup>10</sup> and **Apaf1** (apoptotic protease-activating factor 1)<sup>11</sup>, respectively. On activation, initiator caspases can cleave effector procaspases into their active forms, which are responsible for events such as mitochondrial damage, nuclear membrane breakdown, DNA fragmentation, chromatin condensation and, eventually, cell death.

The Bcl2 family of proteins, which are implicated in the regulation of PCD, comprises members that have either anti-PCD (such as **Bcl2** and **Bcl-x<sub>L</sub>**) or pro-PCD (such as **Bax** and **Bak**) effects<sup>12</sup>. Structurally, they all share some degree of similarity and can have up to four Bcl2-homology domains (BH1–BH4). Besides the many Bcl2 members that contain BH domains, such as Bcl2 *per se* and Bax, there are molecules that share sequence homology only with the BH3 domain, such as **Bid** or **Bim**. These BH3-only proteins can act as intracellular death ligands, proximal to multidomain Bcl2 members, and can connect with proximal signal transduction pathways<sup>13</sup>. Multidomain Bcl2 members can preserve or disrupt mitochondrial integrity by regulating the release of mitochondrial apoptogenic factors such as cytochrome *c*, **Smac/Diablo** or apoptosis-inducing factor (**AIF**). Bcl2 can also inhibit initiator caspases by a mitochondrial-independent mechanism<sup>14</sup>.

**The death receptor (or extrinsic) PCD pathway.** The extrinsic PCD pathway (FIG. 1) is recruited on activation of cell-surface death receptors such as **Fas/CD95** and the tumour necrosis factor receptor 1 (**TNFR1**). Death-receptor activation is initiated by specific ligands called death activators — the Fas ligand binds to Fas, and **TNF $\alpha$**  binds to TNFR1. On binding, the intracellular 'death domains' on these receptors associate with an adaptor protein that contains 'death effector domains', Fas associates with FADD, and TNFR1 associates with FADD and **TRADD** (TNFR-associated protein with death domain). Adaptor proteins then recruit procaspase 8, leading to its activation. Activated caspase 8 can then activate other caspases, either directly or indirectly, by cleaving Bid (FIG. 1). The extrinsic PCD pathway is especially instrumental in pathological conditions in which inflammation is a prominent feature. Because there is a growing appreciation that inflammation is a



**Figure 1 | Molecular pathways of programmed cell death.** Extracellular signals through cellular death receptors, such as Fas, and intracellular signals, including damage to subcellular constituents or endoplasmic reticulum stress, both trigger genetically programmed pathways of programmed cell death (PCD). The two main PCD pathways result in the activation of downstream executioner caspases and cell death. AIF, apoptosis-inducing factor; Cyt. *c*, cytochrome *c*; FasL, Fas ligand; IAP, inhibitor of apoptosis; tBid, truncated Bid.

feature of neurodegeneration with potential pathogenic significance<sup>15</sup>, targeting the extrinsic PCD pathway in neurodegenerative diseases is a warranted approach. In addition to being activated by the extrinsic pathway, procaspase 8 also seems to be cleaved by a mechanism that depends on the intrinsic pathway<sup>16</sup>.

**The mitochondrial (or intrinsic) PCD pathway.** In the intrinsic PCD pathway, death receptor-independent stimuli can trigger PCD by inducing translocation of pro-PCD molecules (such as Bax) to the mitochondria, with the subsequent release of mitochondrial apoptogenic factors (such as cytochrome *c*) to the cytosol (FIG. 1). Once released from the mitochondria, cytochrome *c* interacts with two other cytosolic protein factors<sup>11</sup>, Apaf1 and procaspase 9, to activate **caspace 3**. The formation of this multimeric Apaf1–cytochrome *c* complex might serve to increase the local concentration of procaspases for intermolecular cleavage, and to set a relatively high threshold of caspase activation so that an occasional leakage of cytochrome *c* will not cause cells to undergo PCD<sup>17</sup>. Smac/Diablo is another mitochondrial intermembrane protein that is released into the cytosol on induction of PCD<sup>18,19</sup>. Once in the cytosol, Smac/Diablo interacts with several inhibitors of apoptosis (IAPs), therefore relieving the inhibitory effect of IAPs on initiator (such as caspase 9) and effector caspases (such as caspase 3)<sup>20,21</sup>. In contrast to cytochrome *c* and Smac/Diablo,

the release of AIF and endonuclease G from the mitochondrial intermembrane space does not lead to caspase activation<sup>22,23</sup>. Under certain death-inducing stimuli, AIF translocates from the mitochondria to the nucleus where it induces caspase-independent, large-scale DNA fragmentation<sup>22</sup>. Similarly, endonuclease G, which is normally involved in the replication of mitochondrial DNA, can translocate to the nucleus on induction of PCD, and can induce fragmentation of nuclear DNA<sup>23</sup>. It has been reported that wah-1, the AIF homologue in *Caenorhabditis elegans*, associates and cooperates with endonuclease G to promote DNA degradation and apoptosis<sup>24</sup>.

Stress in the endoplasmic reticulum (ER), including the disruption of calcium homeostasis and accumulation of unfolded proteins in the ER, can also result in PCD<sup>25</sup> through activation of **caspace 12**. Active caspase 12 can, in turn, cleave caspase 9 (REF. 26). Pro-PCD members of the Bcl2 family — such as Bax and Bak — operate at the ER to maintain calcium homeostasis and regulate ER-dependent PCD<sup>27</sup>.

### Targeting PCD in Parkinson's disease

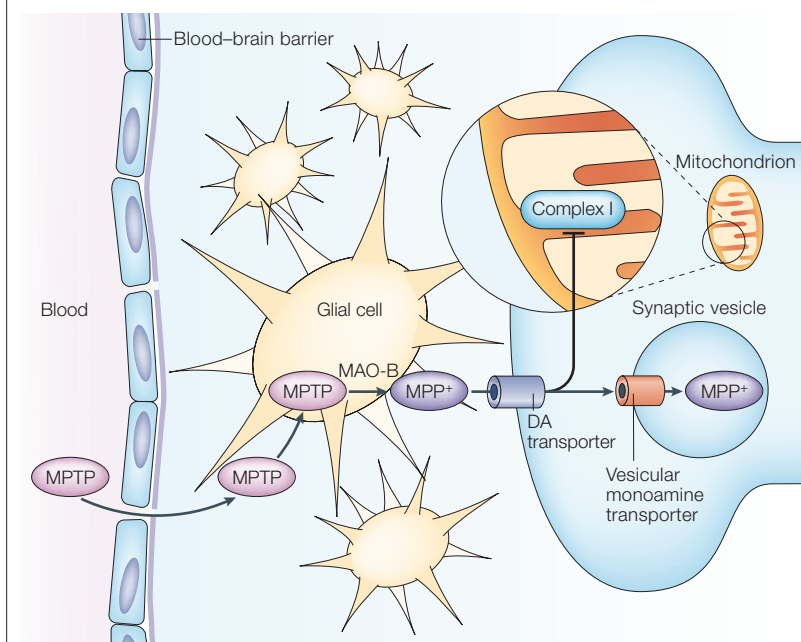
Parkinson's disease (PD) is a common neurodegenerative disorder of unknown cause, characterized by resting tremor, slowness of movement, rigidity and postural instability<sup>28</sup>. PD symptoms are attributed to the loss of dopamine (DA)-containing neurons in the substantia nigra pars compacta (SNPC). In the United States alone, this disease affects about one million people<sup>28</sup>. Although several approved drugs alleviate PD symptoms, none of them seem to stop or slow the neurodegenerative process.

Initially, the demonstration of increased numbers of TUNEL-positive DA neurons in the brains of patients with PD has been used to support the occurrence of apoptosis in this disease<sup>29</sup>. Subsequent studies, using a greater variety of morphological tools, have either succeeded<sup>30–32</sup> or failed<sup>33–35</sup> to find more apoptotic neurons in *post mortem* tissue from PD patients, leading to lively discussions in the field about whether apoptosis in PD is a myth or a reality. Moving away from morphological assessments, immunolocalization of Bax shows that a greater percentage of DA neurons in the SNPC were positive for this pro-PCD protein in brains of patients with PD as compared with controls<sup>36</sup>. In addition, Bax content seemed higher in the remaining DA neurons<sup>37</sup>, consistent with an ongoing neurodegenerative process. DA neurons with increased expression and subcellular redistribution of the anti-PCD protein Bcl-x<sub>L</sub>, and with increased activity of the effector protease caspase 3, have been found in greater abundance in the SNPC of people with PD as compared with controls<sup>38,39</sup>, although these findings could not be independently confirmed<sup>35</sup>. Other PCD-related alterations detected in the brain of patients with PD include the activation of caspase 8 (REFS 40,41) and caspase 9 (REF. 41). But despite this body of descriptive data, the evidence cannot be regarded as an unequivocal demonstration that PCD has a pathogenic role in PD.

To address this crucial issue, it is necessary to use experimental models that allow manipulations of the PCD machinery and to assess their impact on neuronal

## Box 2 | The MPTP model of Parkinson's disease

1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) is a by-product of the chemical synthesis of a meperidine analogue with potent heroin-like effects that can induce a parkinsonian syndrome in humans almost indistinguishable from Parkinson's disease (PD)<sup>154</sup>. Since the discovery that MPTP causes parkinsonism in humans and non-human primates, as well as in various other mammalian species, it has been used extensively as a model of PD. From neuropathological data, MPTP administration causes damage to the nigrostriatal dopamine (DA) pathway identical to that seen in PD, with the exception of the intraneuronal inclusions known as Lewy bodies. It is worthwhile noting that *post mortem* brain samples from patients with PD show a selective defect in the mitochondrial electron transport chain complex that is affected by MPTP<sup>155,156</sup>. The metabolism of MPTP is a complex, multistep process (see figure). After its systemic administration, MPTP, which is a pro-toxin, rapidly crosses the blood–brain barrier and is metabolized to 1-methyl-4-phenyl-2,3-dihydropyridinium (MPDP<sup>+</sup>) by the enzyme monoamine oxidase B (MAO-B) in non-DA cells, and then, probably by spontaneous oxidation, to 1-methyl-4-phenylpyridinium (MPP<sup>+</sup>), the active toxic compound. MPP<sup>+</sup> is then taken up by DA transporters, for which it has high affinity. Once inside DA neurons, MPP<sup>+</sup> is concentrated by an active process within the mitochondria, where it impairs mitochondrial respiration by inhibiting complex I of the electron transport chain. The inhibition of complex I impedes the flow of electrons along the mitochondrial electron transport chain, resulting in an increased production of free radicals, which causes oxidative stress and activation of programmed cell death molecular pathways.



death. For the past twenty years, the 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) mouse model of PD has generated important insights into PD pathogenesis and the role of PCD in PD (BOX 2). In addition to the MPTP model, the chronic infusion of rotenone to rats, which recapitulates most of the pathological hallmarks of PD<sup>42</sup>, is emerging as an invaluable model to investigate PD pathogenesis. But compared to MPTP, the rotenone model is technically challenging and might not be as specific for the SNPC DA neurons<sup>43</sup> as initially claimed<sup>42</sup>. Genetic models of PD, based on the overexpression of  $\alpha$ -synuclein, have also been developed (see later discussion)<sup>44–48</sup>, but the lack of consistent SNPC DA neurodegeneration makes these models more suitable for studying pathogenic processes other than DA neuronal death.

## DOMINANT NEGATIVE

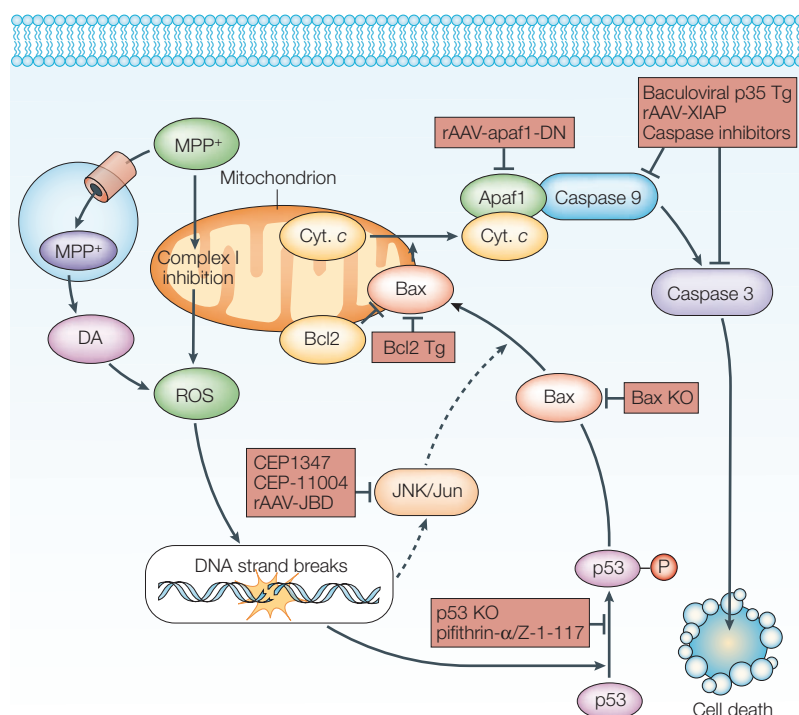
A mutant molecule that can form a heteromeric complex with the normal molecule, knocking out the activity of the entire complex.

Alterations in energy metabolism, generation of reactive oxygen species (ROS) and perturbations in calcium homeostasis occur within hours of MPTP administration — well before any significant neuronal death has occurred<sup>49</sup> — raising the possibility that these events are not the main effectors of cell death. Instead, they could be intracellular signals that can set into motion deleterious molecular cascades — such as PCD activation — which are ultimately responsible for the demise of DA neurons. The prolonged administration of relatively low doses of MPTP to mice leads to morphologically defined apoptotic DA neurons<sup>50</sup>. Under this regime of MPTP intoxication, Bax is strongly upregulated in SNPC DA neurons<sup>51</sup>, whereas Bcl2 is downregulated. In this model, activation of Bax induces the recruitment of the mitochondrial PCD pathway, with the subsequent activation of caspase 9 and caspase 3 (REFS 38,41). The key role of Bax in MPTP-induced neurotoxicity is illustrated by the demonstration that mutant mice deficient in Bax are resistant to the toxicity of MPTP<sup>51</sup>. Overexpression of *Bcl2* also protects DA cells against MPTP-induced neurodegeneration<sup>52,53</sup>.

How the deregulation of Bcl2 family members occurs after MPTP administration is unknown. It is improbable that MPTP directly alters Bax expression and conformation. Instead, it is more plausible that MPTP activates intracellular signalling pathways, which, in turn, cause Bax upregulation and its post-translational activation (for example, Bax oligomerization and internalization into mitochondrial membranes). The tumour suppressor protein p53 is one of the rare molecules known to regulate *Bax* expression<sup>54</sup>, and p53 is activated after MPTP intoxication<sup>55</sup>, probably in response to MPTP-induced DNA damage<sup>56</sup>. Inhibition of p53 attenuates MPTP-induced Bax upregulation and the degeneration of DA neurons<sup>57</sup>. In addition, p53 null mice are resistant to the MPTP-induced death of DA neurons<sup>58</sup>. Activation of the Jun N-terminal kinase (JNK) pathway has also been observed after MPTP administration<sup>59,60</sup>. Moreover, pharmacological blockade of JNK activation with CEP-1347/KT-7515 (REF. 61) or CEP-11004, or its inhibition by adenoviral gene transfer of the JNK-binding domain of JNK-interacting protein 1 (REF. 60) resulted in a marked attenuation of MPTP-induced neurodegeneration. *In vitro* evidence indicates that JNK activation caused by DNA damage is required for the mitochondrial translocation of Bax and the resulting recruitment of the mitochondrial PCD pathway<sup>62,63</sup>. These data indicate that, in the MPTP mouse model of PD, both p53 and JNK might act in concert to cause Bax induction and the post-translational changes that are mandatory to its pro-PCD role.

The intrinsic PCD pathway is recruited after MPTP administration<sup>41</sup>, and blockade of this PCD pathway by an intrastriatal injection of an adeno-associated viral (AAV) vector containing a DOMINANT-NEGATIVE form of Apaf1 prevents the MPTP-induced activation of caspase 3 and the ensuing SNPC neuronal death<sup>64</sup>. By contrast, data on the recruitment and the importance of the extrinsic PCD pathway in the MPTP mouse model is still lacking.





**Figure 2 | Proposed mechanism of MPTP-induced programmed cell death.** 1-methyl-4-phenylpyridinium (MPP<sup>+</sup>), the toxic metabolite of 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP), impairs mitochondrial respiration in dopaminergic neurons by inhibiting complex I of the electron transport chain. Inhibition of complex I impedes the flow of electrons along the mitochondrial electron transport chain, resulting in an increased production of reactive oxygen species (ROS). MPP<sup>+</sup> can also redistribute vesicular dopamine (DA) to the cytosol. There, DA readily auto-oxidizes, thereby generating ROS. Both mitochondrial and cytosolic MPP<sup>+</sup>-related ROS productions damage cellular elements, including DNA, and probably alter the expression of redox-sensitive transcription factors. ROS and damaged DNA activate p53, which induces upregulation of Bax. Bax is subsequently translocated into the mitochondria, probably by mediation of Jun N-terminal kinase (JNK), where it induces the release of cytochrome (cyt.) c to the cytosol and the ensuing caspase activation and cell death. Approaches aimed at targeting different key elements of this cascade (red boxes) result in an attenuation of MPTP-induced neurodegeneration. DN, dominant-negative; JBD, JNK-binding domain; KO, knockout; rAAV, adeno-associated virus vector delivery; tBid, truncated bid; Tg, transgenic; XIAP, X-chromosome-linked inhibitor of apoptosis.

Approaches that are aimed at inhibiting PCD at the level of the effector caspases have given inconsistent results. Adenoviral gene transfer of X-chromosome-linked inhibitor of apoptosis (XIAP) — a protein caspase inhibitor — prevents MPTP-induced SNPC neuronal death, but does not prevent the loss of striatal DA nerve fibres<sup>65</sup>. By contrast, transgenic mice overexpressing p35 — a general caspase inhibitor — in neurons showed attenuated MPTP-induced DA cell death and striatal DA depletion<sup>41</sup>. The discrepancy between these two studies might be explained by the more comprehensive inhibition of caspases mediated by p35 compared with XIAP, as p35 not only inhibits executioner caspases (such as caspases 3 and 7), but also upstream initiator caspases, such as caspases 8 and 9. In both studies, however, DA cell death was assessed early after MPTP administration, raising the question of whether the obtained effect indicates an actual protection or merely a delay in the cell death process. Similar to the findings with XIAP, some *in vitro* studies indicate that it is possible to produce resistance to cell death

selectively at the level of the cell body. The broad-spectrum caspase inhibitor benzylloxycarbonyl-Val-Ala-Asp-fluoromethylketone (zVAD-fmk), and peptide inhibitors of caspases 2, 3 and 9 attenuate the loss of DA-containing ventral midbrain cell bodies (but not neurites) exposed to the active metabolite of MPTP — 1-methyl-4-phenylpyridinium (MPP<sup>+</sup>) — in culture<sup>66</sup>. This observation indicates that the molecular pathways that lead to the destruction of the soma might differ from those that govern axonal degeneration<sup>67</sup>. In another study<sup>40</sup>, zVAD-fmk failed to protect cultured ventral midbrain neurons from the effect of MPP<sup>+</sup>, but instead triggered a morphological switch from apoptosis to necrosis. Minocycline, a clinically approved tetracycline derivative that inhibits microglial activation and presumably also inhibits caspase 1 and caspase 3 (REF. 68), mitigates MPTP-induced neurodegeneration<sup>69,70</sup>. Caspase 8, an initiator caspase in the extrinsic PCD pathway, is also activated in MPTP-treated mice and in patients with PD<sup>40,41</sup>. Paradoxically, activation of caspase 8 in the MPTP model occurs after the recruitment of the intrinsic, and not the extrinsic, PCD pathway<sup>41</sup>, indicating that caspase 8 might participate in the amplification, rather than in the initiation of the PCD in SNPC DA neurons.

Another molecule that has been linked to neurodegeneration in PD is  $\alpha$ -synuclein, mutations of which cause a familial form of PD<sup>71,72</sup>.  $\alpha$ -Synuclein is also an important component of the neuropathological hallmark of the disease — the intracellular inclusions known as Lewy bodies<sup>73</sup>. In MPTP-treated animals,  $\alpha$ -synuclein accumulates and becomes nitrated in the cytosol of SNPC DA neurons<sup>74,75</sup>. Ablation of  $\alpha$ -synuclein in mice prevents MPTP-induced neurodegeneration<sup>76</sup>. Although a direct link between  $\alpha$ -synuclein and DA cell death is not yet well established, expression of mutant  $\alpha$ -synuclein in cell cultures promotes apoptosis<sup>77</sup>, and cytochrome *c* stimulates the aggregation of  $\alpha$ -synuclein *in vitro*<sup>78,79</sup>. Other pathways might also be involved in triggering PCD in PD. For example, in the substantia nigra from patients with PD, the levels of the antioxidant glutathione are reduced<sup>80</sup> — an event that can induce PCD in neurons<sup>81</sup>.

Together, these data indicate that the molecular pathways of PCD are involved in the death of SNPC DA neurons in experimental models of PD (FIG. 2). They support the contention that alterations in the components of the PCD machinery that have been identified in the brain of people with PD are of pathological importance. It is noteworthy that targeting PCD upstream of its execution phase results in a marked attenuation of neurodegeneration, whereas interfering at a more downstream level, such as caspase activation, produces variable results. This is of therapeutic relevance because, once the caspase executioner program is in place, its inhibition might only delay cell death. As the symptoms of PD are caused by the loss of DA terminals in the striatum, preventing the death of SNPC cell bodies without preventing the degeneration of their axons is unlikely to be a helpful therapeutic strategy. Neurons might have two self-destruction programmes, one for PCD in the cell body and a second

### Box 3 | The transgenic *SOD1* model of amyotrophic lateral sclerosis

After the discovery that mutations in the gene that encodes the cytosolic free radical scavenging enzyme copper/zinc superoxide dismutase (*SOD1*) are responsible for the familial form of amyotrophic lateral sclerosis (ALS)<sup>157,158</sup>, it has been shown that the transgenic expression of different *SOD1* mutants in mice<sup>83–85</sup> and rats<sup>86</sup> replicates the clinical and pathological hallmarks of ALS. The age at onset of symptoms and the life span of these transgenic animals varies across different lines, depending on the mutation and its level of expression, but when symptoms appear, they invariably include motor abnormalities that progress with the same pattern as ALS<sup>84,159</sup>. The first motor abnormality in mice is a fine tremor, in at least one limb, when the animal is held in the air by the tail<sup>159</sup>. Thereafter, weakness and atrophy of proximal muscles, predominating in the hindlimbs, develop progressively. At the end-stage of the disease (~140 days), mice transgenic for mutant *SOD1* (G93A) are severely paralyzed and can no longer feed or drink on their own<sup>159</sup>. These end-stage mice have a profound loss of motor neurons (~50%), many dystrophic neurites, marked gliosis, some Lewy body-like intracellular inclusions and motor neurons filled with phosphorylated neurofilaments<sup>83,159–162</sup>.

one for the axon<sup>67</sup>. So, a combination of anti-PCD strategies might be required to obtain optimal clinical benefit from such neuroprotective approaches.

#### Targeting PCD in amyotrophic lateral sclerosis

Amyotrophic lateral sclerosis (ALS) is a relentless, fatal disorder that is characterized by a loss of the motor neurons in the cerebral cortex and spinal cord<sup>82</sup>. The progressive decline of muscular function results in paralysis, speech and swallowing disabilities, emotional disturbance and, ultimately, respiratory failure, causing death within 2–5 years after the onset of the disease. So far, only a few approved treatments (such as mechanical ventilation and riluzole) prolong survival in patients with ALS to some extent.

Whereas about 90% of ALS cases are sporadic (they show no genetic linkage), about 10% are inherited<sup>82</sup>. Of these, about a fifth carry a dominant mutation in superoxide dismutase 1 (*SOD1*). Transgenic expression of different human ALS-linked *SOD1* mutations in mice<sup>83–85</sup> and rats<sup>86</sup> replicates the clinical and pathological hallmarks of ALS (BOX 3), regardless of whether the free-radical scavenging activity of *SOD1* is increased, normal or almost absent<sup>83–87</sup>. This observation, combined with the fact that mutant mice deficient in *SOD1* do not develop any motor neuron disease<sup>88</sup>, indicates that the cytotoxicity of mutant *SOD1* is a GAIN-OF-FUNCTION effect<sup>89</sup>. Transfected neuronal cells that express mutant *SOD1* cDNA undergo apoptosis<sup>90</sup>, as do PC-12 cells transfected with mutant *SOD1* (REF. 91) and primary neurons from transgenic mice that express mutant *SOD1* (REF. 92). These *in vitro* data led investigators to consider whether mutant *SOD1* kills motor neurons by activating PCD. However, inhibition of PCD in transgenic mice that express mutant *SOD1* delays, but does not permanently prevent, neurodegeneration<sup>93,94</sup>. This indicates that the recruitment of PCD in this experimental model of ALS does not result from a direct effect, but from an indirect effect of the mutant protein on the PCD machinery<sup>95</sup>. In keeping with this view, mutant *SOD1* has the propensity to form intracellular aggregates, the presence of which, in the cytosol of motor neurons, might impair microtubule-dependent

axonal transport<sup>96</sup>. It might therefore be possible that mutant *SOD1*, by stimulating protein aggregation, causes important motor neuron perturbations that, in turn, trigger PCD.

Although there are numerous reports on the neuropathological changes in the spinal cord of people with ALS, only a handful provide fine morphological descriptions of the dying motor neurons<sup>97,98</sup>. Some degenerating neurons have some features reminiscent of apoptosis, but none of these dying neurons can confidently be labelled as apoptotic. In mice transgenic for mutant *SOD1*, clear apoptotic cells are seen in the spinal cord, but are they rare<sup>99</sup> and can be immunoreactive for neurofilament or glial fibrillary acid protein<sup>100</sup>, indicating that both neuronal and glial cells are dying by apoptosis in this model. Non-apoptotic PCD variants, such as PARAPTOSIS<sup>3</sup> also seem to occur in this animal model of ALS<sup>101</sup>. The search for other markers of apoptosis in ALS has generated conflicting results. In one *post mortem* study, DNA fragmentation was detected by TUNEL in the spinal cord motor neurons from patients with ALS, but not from controls<sup>102</sup>. In two other studies, DNA fragmentation was detected not only in the motor cortex and spinal cord of people with ALS, but also in controls, although to a lesser degree<sup>103,104</sup>. In a subsequent study, internucleosomal DNA fragmentation was detected in the motor cortex and spinal cord of patients with ALS, but not in spared regions, such as the somatosensory cortex<sup>98</sup>. Here, DNA fragmentation was documented in the anterior horn of the spinal cord and in the motor cortex of patients by gel electrophoresis<sup>98</sup> — a highly specific technique that is not frequently used to identify apoptosis in the nervous system because of its lack of sensitivity. In contrast to these positive findings, other investigators, using similar techniques and tissue samples, have failed to provide any evidence of internucleosomal cleavage of DNA in *post mortem* tissue from patients with ALS or from animal models of the disease<sup>105,106</sup>. Two other apoptotic markers, the Le<sup>y</sup> antigen and fractin, are highly expressed in the spinal cord of people with ALS<sup>102</sup> and transgenic *SOD1* mice<sup>99</sup>, respectively, but neither marker has been detected in control subjects. Likewise, the levels of the PCD-related protein Par-4 (prostate apoptosis response-4) were increased in the spinal cord of patients with ALS and in mice transgenic for mutant *SOD1*, compared with their respective controls<sup>107</sup>.

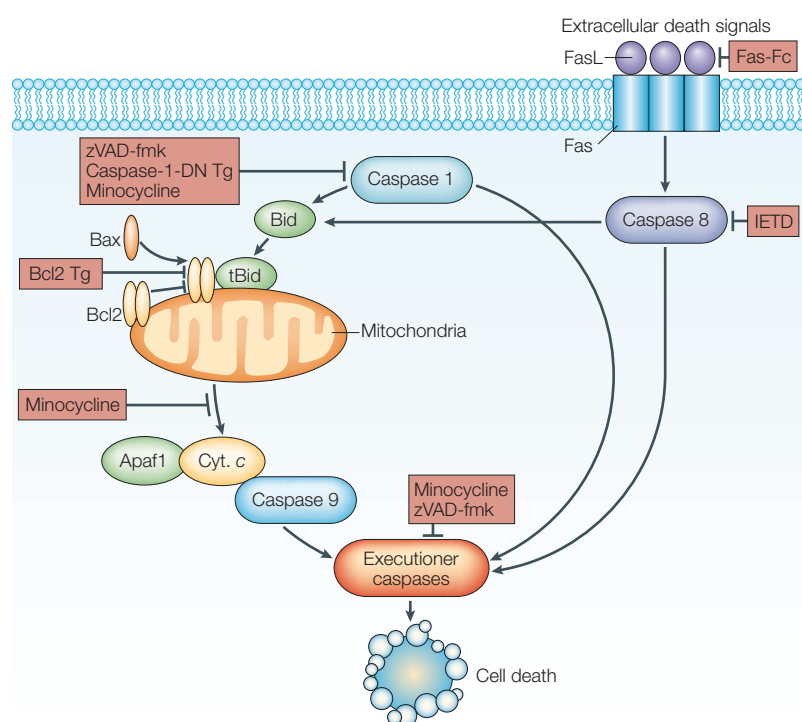
The study of known molecular mediators of PCD, irrespective of the morphology of the dying cells, in *post mortem* ALS samples and in mice transgenic for mutant *SOD1* has provided more consistent results. In the lumbar cord of patients with ALS and transgenic *SOD1* mice, the mRNA content of *Bcl2* is decreased, whereas the level of *Bax* mRNA is increased as compared with controls<sup>108,109</sup>. At the protein level, spinal expression of *Bcl2* and *Bcl-x<sub>L</sub>* is either unchanged<sup>110</sup> or decreased<sup>98,109</sup>, whereas that of *Bax* is increased<sup>98,103,109</sup> both in patients and *SOD1* mice. Because different *SOD1* mutations do not cause exactly the same neuropathology, it is important to stress that a similar pattern of changes of pro- and anti-PCD *Bcl2* family members has been found in the

#### GAIN-OF-FUNCTION

A mutation that confers either a previously inexistent activity to the affected protein or increases a pre-existing function.

#### PARAPTOSIS

A form of programmed cell death that is related to apoptosis. It is transcription dependent and features swelling of the endoplasmic reticulum and mitochondria. However, it does not depend on caspase activation, except for caspase 9, lacks internucleosomal DNA cleavage, and does not show other morphological hallmarks of apoptosis such as nuclear fragmentation.



**Figure 3 | Targeting programmed cell death in amyotrophic lateral sclerosis.** In a transgenic mouse model of amyotrophic lateral sclerosis (ALS), initiator caspases activate downstream effector caspases mainly through the recruitment of the mitochondrial programmed cell death pathway. Interfering with different molecular elements of this cascade (red boxes), and particularly with caspase activation, significantly delays disease onset and mortality in this experimental model of ALS. Cyt., cytochrome; DN, dominant negative; FasL, Fas ligand; IETD Ile-Glu-Thr-Asp-fluoromethylketone; tBid, truncated Bid; Tg, transgenic; zVAD-fmk, benzyloxycarbonyl-Val-Ala-Asp-fluoromethylketone.

spinal cords of another line of mice transgenic for SOD1 that harbour a different SOD1 mutation<sup>111</sup>. None of these alterations, however, are seen in young asymptomatic transgenic mice, but they become progressively more manifest as the neurodegenerative process evolves<sup>109</sup>. In the spinal cord of both patients and affected transgenic SOD1 mice, Bax is not only upregulated, but it is also mainly expressed in its deleterious homodimeric conformation<sup>98,109</sup>. In mice, Bax is relocated from the cytosol to the mitochondria<sup>98,112</sup>. These data indicate that, in ALS, the finely tuned balance between cell death antagonists and agonists is skewed towards death. Consistent with this idea, overexpression of Bcl2 mitigates neurodegeneration in both *in vitro* and *in vivo* models of ALS<sup>91,93</sup> and prolongs survival in transgenic SOD1 mice<sup>93</sup>. As in the MPTP model, Bax upregulation in ALS is associated with p53 activation<sup>111,113</sup>. However, in contrast to the MPTP model, targeting p53 did not confer any protection against mutant SOD1-mediated neurodegeneration<sup>114,115</sup>.

The pro-PCD protein Bid is expressed in the spinal cord of transgenic SOD1 mice and is cleaved into its most active form during the progression of the disease<sup>116</sup>. However, whereas cleaved Bid is seen in mice transgenic for mutant SOD1 as early as the beginning of symptoms, activated caspase 8, which is known to cleave Bid, is detected only at the end-stage of the disease<sup>116</sup>.

This observation indicates that cleavage of Bid might occur in this ALS model by another mechanism, perhaps involving caspase 1 (REF. 116), the inhibition of which prolongs the life-span of these animals<sup>117</sup>. Evidence for a prominent recruitment of the mitochondrial PCD pathway (that is, release of cytochrome *c* from the mitochondria to the cytosol) has been found in the spinal cord of patients and transgenic SOD1 mice<sup>112</sup>. Moreover, pharmacological inhibition of cytochrome *c* release delays disease onset and extends survival of transgenic SOD1 mice<sup>118</sup>. Activation of effector caspases, such as caspases 3 and 7, has also been reported in the spinal cord of patients with ALS<sup>98</sup> and of transgenic SOD1 mice in a time-dependent manner that parallels the progression of the neurodegenerative process<sup>99,100</sup>. An instrumental role for caspases in ALS-related neurodegeneration is evidenced by the demonstration that zVAD-fmk attenuates mutant SOD1-mediated cell death in transfected PC-12 cells<sup>91</sup> and in transgenic SOD1 mice<sup>94</sup>, resulting in a delayed disease onset and mortality in these animals<sup>94</sup>. It has also been reported that mRNA and protein levels of the intrinsic caspase inhibitor XIAP are decreased in the spinal cord of symptomatic SOD1 mice, and that induction of XIAP *in vitro* rescues cells harbouring mutant SOD1 from death by inhibiting the activation of caspase 3 (REF. 119). Activation of caspase 8, which mediates the death receptor pathway triggered by Fas, has been implicated in ALS in cultured embryonic motor neurons<sup>120</sup>. In this study, apoptotic cell death induced by trophic factor deprivation was rescued by Fas-Fc, an antibody fragment that blocks interactions between Fas and FasL, and by the caspase-8 inhibitor Ile-Glu-Thr-Asp-fluoromethylketone (IETD-fmk)<sup>120</sup>. In the presence of neurotrophic factors, exogenous Fas activators such as soluble FasL or anti-Fas antibodies triggered apoptosis of half of the purified motor neurons that were blocked by IETD-fmk<sup>120</sup>. Motor neurons isolated from transgenic mice that overexpress different ALS-linked SOD1 mutants showed increased susceptibility to activation of the Fas-triggered death pathway<sup>121</sup>. However, *in vivo* studies in mice transgenic for mutant SOD1 indicate that activation of caspase 8, like induction of TNF $\alpha$  (REF. 122), occurs in the spinal cord only near end-stages of the cell-death process<sup>116</sup>. This observation indicates that the extrinsic PCD pathway in this model might make a late contribution to the neurodegenerative process.

Together (FIG. 3), the data that we have reviewed here indicate that interfering with PCD delays neuronal death and prolongs survival in experimental models of ALS. Eventually, these mice die despite the different interventions that we have discussed, indicating that targeting the PCD cascade can slow the death process but cannot abrogate it.

#### Targeting PCD in Huntington's disease

Huntington's disease (HD) is an autosomal dominant neurodegenerative disorder characterized by choreic involuntary movements that result from selective neuronal loss in the striatum and the cerebral cortex. HD is caused by expansions of the trinucleotide CAG in

the *huntingtin* gene, producing a protein containing polyglutamine repeats<sup>123</sup>. Transgenic mice that overexpress a fragment of human huntingtin with an extended polyglutamine region show reduced survival, intraneuronal aggregates and behavioural deficits similar to HD<sup>124</sup>. Of these transgenic mice lines, the R6/2 line has been the most thoroughly studied. The length of the polyglutamine repeat is inversely correlated with the age of disease onset. HD is fatal, with a mean survival after onset of 15–20 years, and, at present, there is no effective treatment.

Although it is not known how mutant huntingtin promotes cell death, a self-amplification cascade of caspase activation has been proposed to participate. In *post mortem* brain tissue from patients with HD, TUNEL-positive cells, the morphology of which is reminiscent of what is found in apoptosis, have been detected in the neostriatum<sup>125,126</sup>. There is also evidence of the activation of caspases 1 and 8 (REFS 127,128) in the brain of patients with HD. Activation of caspase 1 was assessed by determining the levels of one of its substrates, interleukin-1 $\beta$ , and this might therefore reflect an inflammatory response rather than an activation of PCD pathways. Also, it is not clear whether activation of caspase 1 occurred in neurons or other cell types.

In experimental models of HD, there is evidence for the activation of caspases 1 and 3 (REF 127). Furthermore, administration of ZVAD-fmk<sup>127</sup> or minocycline<sup>68</sup> — which inhibit caspases 1 and 3 — delays disease progression and mortality in the R6/2 transgenic mouse model of HD. Expression of a dominant-negative mutant form of caspase 1 in R6/2 mice extends survival and delays the appearance of neuronal inclusions, receptor alterations and the onset of symptoms<sup>127</sup>. Transgenic mice that express the caspase 1 dominant-negative mutation are also more resistant to neurotoxins that have been used to model HD, such as malonate and 3-nitropropionic

acid<sup>129</sup>. Dietary restriction suppresses activation of caspase 1, reduces brain atrophy and the formation of huntingtin aggregates, delays disease onset and increases survival in another transgenic mouse model of HD, produced by the expression of a human amino-terminal truncated huntingtin with 82 polyglutamine repeats (HD-N171-82Q)<sup>130</sup>.

Although these findings indicate that caspases might be valuable targets for therapeutic intervention in HD, how mutant huntingtin triggers PCD remains an enigma. It has been shown<sup>131</sup> that expression of extended polyglutamine repeats *in vitro* can directly activate initiator and effector caspases, such as caspases 3, 8 and 9. Moreover, wild-type huntingtin can attenuate PCD in cultured cells by preventing activation of caspase 3 through inhibition of the processing and activity of caspase 9 (REF 132). However, the relationship between expanded polyglutamine repeats and PCD pathways seems more complex because, in addition to triggering the activation of PCD, some polyglutamine-containing proteins are caspase substrates. It has been shown that caspase 3 can cleave wild-type and mutant huntingtin proteins *in vitro*, thereby generating truncated fragments<sup>133–135</sup>. Truncated fragments that contain expanded polyglutamine repeats show increased toxicity and propensity to aggregate, compared with the full-length protein<sup>133,134,136</sup>. Huntingtin fragments have been identified in the brain of people with HD<sup>137–140</sup>, but also in controls, indicating that caspase-mediated cleavage of wild-type huntingtin might occur as a normal physiological event. However, it is thought that cleavage of mutant huntingtin would release fragments with the potential for increased toxicity and accumulation, owing to the presence of the expanded polyglutamine tract<sup>140</sup>.

Caspase 8 is another caspase that might be involved in the pathogenesis of HD. This caspase is recruited to

Table 1 | *In vivo* and *in vitro* approaches to attenuate neurodegeneration in experimental models by targeting programmed cell death

| Disease | Experimental model                            | Genetic manipulation                             | Viral vector delivery                                   | Pharmacological inhibition                          |
|---------|-----------------------------------------------|--------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------|
| PD      | MPTP-treated mice                             | Bax KO <sup>51</sup> of JIP-1 (REF. 60)          | JNK binding domain CEP-1347/KT-7515 (JNK) <sup>61</sup> | Pifithrin- $\alpha$ /Z-1-117 (p53) <sup>57</sup>    |
|         |                                               | Bcl2 Tg <sup>52,53</sup>                         |                                                         | CEP-11004 (JNK) <sup>163</sup>                      |
|         |                                               | p53 KO <sup>58</sup>                             | Apaf1 (REF. 64)                                         | Minocycline (caspases1, 3, cyt. c) <sup>69,70</sup> |
|         |                                               | Baculoviral p35 Tg <sup>41</sup>                 | XIAP <sup>65</sup>                                      |                                                     |
| ALS     | Mesencephalic dopaminergic neurons in culture |                                                  |                                                         | zVAD-fmk/M-920 (REF. 66)                            |
|         |                                               |                                                  |                                                         | zVDVAD-fmk (caspase 2) <sup>66</sup>                |
|         |                                               |                                                  |                                                         | zDEVD-fmk/M-725 (caspase 3) <sup>66</sup>           |
| ALS     | Transgenic <i>SOD1</i> mice                   | Bcl2 Tg <sup>93</sup>                            |                                                         | zLEHD-fmk (caspase 9) <sup>66</sup>                 |
|         |                                               | Caspase 1-DN Tg <sup>117</sup>                   |                                                         | zVAD-fmk <sup>94</sup>                              |
|         |                                               |                                                  |                                                         | Minocycline (caspase 1, 3, cyt. c) <sup>118</sup>   |
| HD      | Embryonic motor neurons in culture            |                                                  |                                                         | IETD-fmk (caspase 8) <sup>120</sup>                 |
|         |                                               |                                                  |                                                         | Fas-Fc (Fas-FasL) <sup>120</sup>                    |
|         |                                               | Caspase 1-DN Tg <sup>127</sup>                   |                                                         | zVAD-fmk <sup>127</sup>                             |
| HD      | Transgenic R6/2 mice                          |                                                  |                                                         | Minocycline (caspase 1, 3, cyt. c) <sup>68</sup>    |
|         |                                               | Wild-type huntingtin transfection <sup>132</sup> |                                                         |                                                     |

ALS, amyotrophic lateral sclerosis; cyt., cytochrome; DN, dominant-negative; FasL, Fas Ligand; HD, Huntington's disease; IETD, Ile-Glu-Thr-Asp-fluoromethylketone; JNK, Jun N-terminal kinase; JIP, JNK interacting protein-1; KO, knockout; MPTP, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; PD, Parkinson's disease; SOD1, superoxide dismutase; Tg, transgenic; XIAP, X-chromosome-linked inhibitor of apoptosis; zDEVD-fmk, benzyloxycarbonyl-Asp(OMe)-Glu(OMe)-Val-Asp(OMe)-fluoromethylketone; zLEHD-fmk, benzyloxycarbonyl-Leu-Glu(OMe)-His-Asp(OMe)-fluoromethylketone; zVAD-fmk, benzyloxycarbonyl-Val-Ala-Asp-fluoromethylketone; zVDVAD-fmk, benzyloxycarbonyl-Val-Asp-Val-Ala-Asp-fluoromethylketone.



intracellular aggregates and is subsequently activated in neuronal cells that express an expanded polyglutamine repeat<sup>128</sup>. In this system, caspase inhibition prevents cell death but not inclusion formation<sup>128</sup>. Furthermore, activated caspase 8 has been detected in the insoluble fractions of brains of people with HD but not of controls, further supporting the idea that caspase 8 is recruited to aggregates and subsequently activated<sup>128</sup>. It has also been proposed that caspase 8 activation in HD is mediated by the formation of pro-PCD heterodimers between **Hip1** (huntingtin interacting protein 1) and **Hippi** (Hip1 protein interactor), which is favoured by the disease-associated polyglutamine expansion<sup>141</sup>. This non-receptor-mediated pathway for activating caspase 8 might trigger PCD through components of the extrinsic cell-death pathway and contribute to neuronal death in HD.

Together, these data indicate a complex association between expanded polyglutamine repeat-containing proteins and caspase activation, indicating that caspases might also be a useful target for therapeutic intervention in HD.

### Conclusions and perspectives

The different studies that we have reviewed here indicate that key molecular components of PCD are recruited in PD, ALS and HD. Much of the data on the occurrence of PCD in these diseases has been obtained by studying *post mortem* human brain samples. However, the information on the temporal relationship between these molecular changes, and their importance in the pathologic cascade, emanates largely from the use of animal models. Although not all of the diseases that we have reviewed show identical PCD changes, two general conclusions can be formulated. First, activation of PCD molecular pathways is a consistent feature of neurodegeneration. Second, PCD is not the sole mediator of cell demise in these disorders, but is a key component within a coalition of deleterious mechanisms that are responsible for the degenerative process.

The question of whether PCD pathways are involved in the demise of neurons in neurodegenerative diseases is more than academic, as it has important implications for the rational development of therapeutic strategies

(TABLE 1). However, although the molecular complexity of the PCD cascade potentially offers many opportunities for its modulation, such pathways contain few conventional drug targets, such as enzymes and receptors. Therefore, researchers are focusing on other strategies to affect the components of the PCD pathway<sup>142</sup>. One approach is the modulation of the expression of key molecular components of the PCD machinery by gene and antisense therapy, but these technologies still need further development. Also, the problem of whether PCD can be selectively modulated in a specific organ or cell type, without adverse effects on others, remains to be solved. This issue is especially important in the context of chronic processes such as neurodegeneration that would require a sustained anti-PCD treatment. This problem could eventually be solved by fusing anti-PCD molecules with other molecules that are tagged to be recognized by tissue-specific receptors or uptake systems (for example, DA transporters). Alternatively, the problem could be tackled by synthesizing anti-PCD molecules as inactive pro-molecules that would be enzymatically activated in specific tissues or cell types (for example, by tyrosine hydroxylase in DA neurons).

One of the few potential pharmacological approaches that could modulate PCD involves inhibitors of caspase activity. According to the data that we have reviewed here, caspase inhibition might indeed delay cell death in different experimental models of neurodegeneration. However, targeting PCD at such a downstream point seems insufficient to stop the degenerative process. Nevertheless, it should be possible to combine caspase inhibition with other treatments directed at upstream events, as we believe that activation of PCD in neurodegenerative diseases is not a primary event. Instead, we think it is a 'suicide' decision, taken by cells that are affected by disease-related abnormalities such as inflammation, mitochondrial dysfunction, oxidative stress, or misfolding and aggregation of proteins. Targeting PCD alone is therefore unlikely to be sufficient to obtain a beneficial therapeutic effect. Optimal neuroprotection might require administration of a pharmacological cocktail directed against several pathogenic events.

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